

Linear Algebra Solutions

Claudio Bastiani-Fonck

Mathematics is the art of reducing any problem to linear algebra.

-William Stein

Basic Linear Algebra

1

Notice that the matrix is circulant. Therefore the rank is $(2n+1) - d$ where d is the degree of

$$\gcd(f(x), x^{2n+1} - 1)$$

where $f(x) = x + x^2 + \dots + x^n - x^{n+1} - x^{n+2} - \dots - x^{2n}$ is the accompanying polynomial. Rewrite each block as a geometric sum (for $x \neq 1$):

$$x + x^2 + \dots + x^n = \frac{x(1-x^n)}{1-x}, \quad x^{n+1} + x^{n+2} + \dots + x^{2n} = \frac{x^{n+1}(1-x^n)}{1-x}.$$

Hence

$$f(x) = \frac{(1-x^n)(x-x^{n+1})}{1-x} = \frac{x(1-x^n)^2}{1-x}.$$

Now use $1-x^n = (1-x)(1+x+\dots+x^{n-1})$ to cancel:

$$f(x) = x(1-x)(1+x+\dots+x^{n-1})^2.$$

Therefore the roots are:

$$x = 0, \quad x = 1, \quad \text{and } \omega \text{ with } \omega^n = 1, \omega \neq 1,$$

with multiplicity 2 for each nontrivial n th root of unity (because of the square), and multiplicity 1 for $x = 1$. The only overlap with $2n+1$ roots of unity is 1, therefore the gcd has degree 1 and the rank of the matrix is $2n+1-1=2n$.

Now we will derive the identity. Consider a circulant matrix

$$C = \begin{pmatrix} c_0 & c_{n-1} & c_{n-2} & \cdots & c_2 & c_1 \\ c_1 & c_0 & c_{n-1} & \cdots & c_3 & c_2 \\ c_2 & c_1 & c_0 & \cdots & c_4 & c_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ c_{n-2} & c_{n-3} & c_{n-4} & \cdots & c_0 & c_{n-1} \\ c_{n-1} & c_{n-2} & c_{n-3} & \cdots & c_1 & c_0 \end{pmatrix} \quad f(x) = c_0 + c_1x + \dots$$

and write C as a polynomial in the cyclic permutation matrix P :

$$C = c_0 + c_1P + c_2P^2 + \dots + c_{n-1}P^{n-1}$$

We want to compute the nullity of this expression. Note that P has characteristic polynomial $x^n - 1$ (it is a companion matrix) so it is diagonalizable with distinct eigenvalues ζ_n^i . Considering the expression in this eigenbasis gives the result (0 eigenvalues correspond to shared roots).

2

This is actually true in n dimensions (as the argument can generalize). Consider the matrix equation: Let A be the matrix with entries a_{ij} of the form

$$A = \begin{bmatrix} + & - & - \\ - & + & - \\ - & - & + \end{bmatrix}$$

Note that $tI - A$ is strictly positive for large enough t . Therefore by Perron–Frobenius it has a strictly positive eigenvector v with eigenvalue λ . Then $Av = (t - \lambda)v$ has all entries with the same sign.

3

Consider the matrix

$$A = \begin{bmatrix} y & z & x \\ x & y & z \\ z & x & y \end{bmatrix}$$

taking the adjugate matrix gives:

$$\text{adj } A = \begin{bmatrix} b & a & c \\ c & b & a \\ a & c & b \end{bmatrix}$$

which is expressed only in terms of a and b . Now to work backward recall the linear algebra identities

$$\text{adj}(\text{adj } A) = (\det A)^{n-2} \cdot A \implies \det \text{adj}(\text{adj } A) = \det A^{n-1}$$

where n is the dimension. This gives for $n = 3$

$$A = \frac{\text{adj}(\text{adj } A)}{\pm \sqrt{\det \text{adj } A}}$$

which gives explicit formulas for x, y, z in terms of a, b, c by computing the opposite diagonal terms:

$$x = \frac{a^2 - bc}{\sqrt{a^3 + b^3 + c^3 - 3abc}} \quad y = \frac{b^2 - ac}{\sqrt{a^3 + b^3 + c^3 - 3abc}} \quad z = \frac{c^2 - ab}{\sqrt{a^3 + b^3 + c^3 - 3abc}}$$

4

No it does not follow. Take

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

5

Note that a symplectic matrix is a $2n \times 2n$ matrix M with

$$M^T J M = J \quad J = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix}$$

so named because it corresponds to a symplectic form (alternating bilinear)

$$\omega : V \times V \rightarrow k \quad \omega(-, -) = \langle -, J- \rangle \quad \omega(M-, M-) = \omega(-, -)$$

where J is any skew symmetric matrix (however our choice is standard).

Theorem 1.1 (Symplectic). Consider a block matrix:

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

Then the following are equivalent

- (a) M is symplectic
- (b) $A^T C, B^T D$ are symmetric and $A^T D - C^T B = I$.
- (c) AB^T, CD^T are symmetric and $AD^T - BC^T = I$.

Proof. Compute

$$M^T J M = \begin{bmatrix} A^T C - C^T A & A^T D - C^T B \\ B^T C - D^T A & B^T D - D^T B \end{bmatrix}$$

which immediately proves (1) $\Leftrightarrow$ (2). We can also compute

$$M J M^T = \begin{bmatrix} AB^T - BA^T & AD^T - BC^T \\ CB^T - DA^T & CD^T - DC^T \end{bmatrix}$$

so it suffices to show that M symplectic implies M^T is symplectic (and the backward arrow follows by symmetry). Write:

$$\begin{aligned} M^T J M &= J \\ J M J^{-1} &= M^{-T} \\ J M^T J^{-1} &= M^{-1} && \text{take transpose and use skew symmetry} \\ M J M^T &= J \end{aligned}$$

□

The problem immediately follows from this theorem.

6

Firstly associate to each x_i the projection $x_i x_i^T$. We claim that these projections are linearly independent. Suppose that we have a nontrivial linear relation:

$$\sum_i a_i x_i x_i^T = 0$$

We multiply by x_k^T on the left and x_k on the right to yield

$$\sum_i a_i x_k^T x_i x_i^T x_k = \sum_i a_i \langle x_k, x_i \rangle^2 = a_k + \sum_{i \neq k} a_i \alpha^2 = 0 \quad (1)$$

which gives N equations constraining the coefficients a_i . Summing these equations yields

$$\sum_i a_i (1 + (N-1)\alpha^2) = 0 \implies \sum_i a_i = 0$$

and now subtract α^2 times this from Eq. (1) to yield

$$a_k (1 - \alpha^2) = 0 \implies a_k = 0$$

as desired.

Now that the $x_i x_i^T$ are linearly independent it suffices to show that they exist in a vector space of dimension at most $n(n+1)/2$. Indeed, they are all symmetric matrices and the space of symmetric matrices of size n has dimension $n(n+1)/2$. This is because the entries above and on the diagonal can be chosen and then the rest are determined.

Determinants

7

Subtracting the first column from all the other columns to get

$$\begin{bmatrix} 3 & -2 & -2 & -2 & \cdots & -2 \\ 1 & 3 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 4 & 0 & \cdots & 0 \\ 1 & 0 & 0 & 5 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & 0 & \cdots & n \end{bmatrix}$$

Then subtract multiples of the other columns from the first to kill the left column

$$\begin{bmatrix} 3 + 2\left(\frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n}\right) & -2 & -2 & -2 & \cdots & -2 \\ 0 & 3 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 4 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 5 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & n \end{bmatrix}$$

Pulling in $3 = 2(1 + \frac{1}{2})$ to get a harmonic sum and multiplying along the diagonal gives

$$H_n n!$$

however the harmonic numbers are unbounded so the thing is not bounded.

8

Note that the degree of the determinant is $p^2 + p + 1$, this reminds one of the projective plane $\mathbb{P}_2(\mathbb{F}_p)$. Intuitively the determinant expresses a nontrivial linear relation, write

$$\det \begin{bmatrix} x & y & z \\ x^p & y^p & z^p \\ x^{p^2} & y^{p^2} & z^{p^2} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$$

giving:

$$ax + by + cz = 0 \quad (ax + by + cz)^p = 0 \quad (ax + by + cz)^{p^2} = 0$$

We therefore claim that the determinant is (up to a constant):

$$\det \begin{bmatrix} x & y & z \\ x^p & y^p & z^p \\ x^{p^2} & y^{p^2} & z^{p^2} \end{bmatrix} = \prod_{(a,b,c) \in \mathbb{P}_2(\mathbb{F}_p)} (ax + by + cz)$$

By the UFD property of $\mathbb{F}_p[x, y, z]$, the irreducibility of linear forms, and a degree argument, it suffices to show that each of these terms divide the determinant. To show this rigorously, work in the quotient $\mathbb{F}_p[x, y, z]/(ax + by + cz)$, the previous argument shows that the columns are linearly dependent in this quotient. Therefore the determinant is in the ideal $(ax + by + cz)$ as desired.

Note that this is called the Moore determinant.

9

This is asking to derive the Pfaffian, there are many online sources for a rigorous argument. Here is my way of computing it (by expanding along the top row)

$$\text{Pf}(A) = \sum_{j=2}^{2n} (-1)^j a_{1j} \text{Pf}(A_{\hat{1}\hat{j}})$$

where $A_{\hat{1}\hat{j}}$ is A with the 1, j rows and columns removed. We will use a more canonical definition of the Pfaffian for the proof.

Proof. Let $A = (a_{ij})$ be a $2n \times 2n$ skew-symmetric matrix. We wish to show that $\det(A) = [\text{Pf}(A)]^2$.

Recall that the Pfaffian is defined as a sum over perfect matchings $\mathcal{M}$:

$$\text{Pf}(A) = \sum_{M \in \mathcal{M}} \text{sgn}(M) \prod_{(i,j) \in M} a_{ij}.$$

where if $\mathcal{M} = (i_1, j_1) \cdots (i_n, j_n)$ with $i_k < j_k$ then

$$\text{sgn}(\mathcal{M}) := \text{sgn}([i_1, j_1, i_2, j_2, \dots, i_{2n}, j_{2n}]) \quad (1)$$

and it does not matter how we index the pairs. We need to show that the square of the Pfaffian

$$[\text{Pf}(A)]^2 = \sum_{M, N \in \mathcal{M}} \text{sgn}(M) \text{sgn}(N) \prod_{(i,j) \in M} a_{ij} \prod_{(k,l) \in N} a_{kl}.$$

is the determinant, ie the sum of signed products of paths down the matrix. Consider the graph G formed by superimposing the edges of M and N . Since M and N are perfect matchings, every vertex in G has degree exactly 2. Therefore G decomposes uniquely into a disjoint union of cycles.

Because the edges in any such cycle must alternate between M and N , every cycle must have even length. Let these edges define a permutation $\sigma \in S_{2n}$ such that for every vertex i , the edge $(i, \sigma(i))$ is in $M \cup N$.

This establishes a bijection:

$$\Phi : (M, N) \longleftrightarrow \sigma \in S_{2n} \text{ (where } \sigma \text{ has only even cycles).}$$

For a skew-symmetric matrix, permutations with any odd-length cycles vanish in the determinant expansion (due to the cancellation $a_{ji} = -a_{ij}$ and pairing with the reversed cycle). Thus, the determinant is the sum over exactly these valid σ :

$$\det(A) = \sum_{\substack{\sigma \in S_{2n} \\ \text{even cycles only}}} \text{sgn}(\sigma) \prod_{i=1}^{2n} a_{i, \sigma(i)}.$$

To equate the expressions, we need to show that the signs are matching. IE that:

$$\text{sgn}(M) \text{sgn}(N) = \text{sgn}(\sigma) \cdot (-1)^{\text{number of } i \text{ with } \sigma(i) < i}$$

It can be seen that changing the order of the elements in a cycle changes both sides the same way (try to see what happens to Eq. (1)). Therefore we may assume all the cycles are increasing with one final "down

step". Then the RHS cancels to 1 since each cycle contributes $(-1)^2$. On the LHS each cycle of length $2k$ gives:

$$M = [v_1 v_2 v_3 v_4 \cdots v_{2k-1} v_{2k}] \quad N = [v_2 v_3 v_4 v_5 \cdots v_{2k-2} v_{2k-1} v_1 v_{2k}]$$

and we see M and N have the same sign (takes $2k - 2$ swaps to move v_1 to the right spot), so cancel out. Therefore, the sum over pairs of matchings corresponds exactly to the sum over even-cycle permutations:

$$[\text{Pf}(A)]^2 = \det(A).$$

□

Doing some research on this shows a lot of claimed proofs using some wedge product magic. Many are very convincing, but I can't really claim to be confident they descend to a full proof, there is a lot of reasonable but perhaps nontrivial claims being made. They also aren't very constructive, and I suppose in the math competition setting one should prefer proofs that let you understand the formula. Of course, I have no idea what I'm talking about, I'm just justifying why I wrote this proof (which I like a lot).

10

Note that the determinant has a 0 of at least order 1 when either $x_i = x_j$ or $y_i = y_j$ for $i \neq j$. Furthermore the determinant has a pole of at least order 1 when $x_i = y_j$. This is by noting exactly one entry goes to ∞ and by induction the minor on that cofactor has nonzero determinant (for distinct x_i, y_j). Therefore we know that

$$\frac{\prod_{1 \leq i < j \leq n} (x_i - x_j)(y_j - y_i)}{\prod_{1 \leq i, j \leq n} (x_i - y_j)}$$

divides the determinant. Furthermore by comparing degrees this expression must be the determinant up to a constant. To compute the constant let $y_i = x_i - 1$, $x_1 \gg 1$ and $x_{i+1} \gg x_i$. Therefore the matrix becomes diagonal and has determinant 1. Furthermore the product becomes

$$= \frac{\prod_{1 \leq i < j \leq n} (x_i - x_j)(y_j - y_i)}{\prod_{1 \leq i, j \leq n} (x_i - y_j)} \approx \frac{1}{\prod (x_i - y_i)} \approx 1$$

and the constant is 1.

Recall that the entries $b_{i,j}$ of the inverse matrix H^{-1} are given by the ratio of the cofactor $C_{j,i}$ to the determinant $\det(H)$. We can compute the cofactor $C_{j,i}$ using the exact same formula where x_j and y_i are excluded.

$$b_{i,j} = \frac{C_{j,i}}{\det(H)} = \frac{(-1)^{i+j}}{i+j-1} \prod_{k=1}^n \frac{(x_j - y_k)(x_k - y_i)}{(x_j - x_k)(y_i - y_k)}$$

(where the product excludes $k = j$ and $k = i$ in the denominator).

Now consider the matrix with entries $1/(i+j-1)$. By substituting $x_k = k$ and $y_k = 1-k$, this expression resolves into factorials:

$$b_{i,j} = (-1)^{i+j} (i+j-1) \frac{(n+i-1)!}{(n-j)!(i+j-1)!} \frac{(n+j-1)!}{(n-i)!(i+j-1)!} \left[\frac{(i+j-2)!}{(i-1)!(j-1)!} \right]^2$$

We can express $b_{i,j}$ as a product of binomial coefficients and integers:

$$b_{i,j} = (-1)^{i+j} (i+j-1) \binom{n+i-1}{n-j} \binom{n+j-1}{n-i} \binom{i+j-2}{i-1}^2$$

As desired.

Note that this is called the Cauchy matrix. The Hilbert matrix is the one with entries $1/(i+j-1)$ and is also a special form of a Hankel matrix, which appears later.

11

We would like to compute the eigenvalues of A . At this point we can add 1 to all of them to get the eigenvalues of $I + A$ and multiply them all together. We may write $\omega = e^{2\pi i/n}$ as the n th root of unity. Therefore

$$A_{jk} = \frac{\omega^{j+k} + \omega^{-j-k}}{2}$$

so we may write

$$u = \begin{bmatrix} \omega \\ \omega^2 \\ \vdots \\ \omega^n \end{bmatrix} \quad v = \begin{bmatrix} \omega^{-1} \\ \omega^{-2} \\ \vdots \\ \omega^{-n} \end{bmatrix}$$

and $A = \frac{1}{2}(uu^T + vv^T)$. Note that both $\frac{1}{2}uu^T$ and $\frac{1}{2}vv^T$ have rank at most 1. therefore A has rank at most 2 and its eigenvalues are in the subspace $\text{Span}(u, v)$. Furthermore note that

$$u^T u = \lambda_u = \omega^2 + \omega^4 + \dots + \omega^{2n} = \frac{\omega^2 - \omega^{2n+2}}{1 - \omega^2} = 0 \quad v^T v = \omega^{-2} + \omega^{-4} + \dots + \omega^{-2n} = \frac{\omega^2 - \omega^{-2n-2}}{1 - \omega^{-2}} = 0$$

$$u^T v = 1 + 1 + \dots + 1 = n \quad v^T u = 1 + 1 + \dots + 1 = n$$

therefore

$$Au = \frac{1}{2}nv \quad Av = \frac{1}{2}nu$$

so that the eigenvalues are $u+v$ and $u-v$ with eigenvalues $\frac{n}{2}$ and $-\frac{n}{2}$ respectively. Therefore the determinant of $A + I$ is

$$\left(1 + \frac{n}{2}\right) \left(1 - \frac{n}{2}\right) = 1 - \frac{n^2}{4}$$

It is worth noting that this only works for $n > 2$ due to the denominator in our formulas for $u^T u$ and $v^T v$. The cases of $n = 1$ and $n = 2$ must be computed separately (and the answers don't agree with our formula).

12

Note that $B = A - A^t$ can be generated by the following probabilities.

$$B_{ij}, B_{ji} = \begin{cases} i = j, & 0, 0 \\ i \neq j, & \begin{cases} 0, 0 \text{ with probability } 1/2 \\ 1, -1 \text{ with probability } 1/4 \\ -1, 1 \text{ with probability } 1/4 \end{cases} \end{cases}$$

Now note that

$$\det B = \sum_{\sigma \in S_{2n}} \text{sign}(\sigma) \prod_i B_{i\sigma(i)}$$

so by linearity of expectation we would like to compute the expected value of each path $\prod_i B_{i\sigma(i)}$. Suppose that σ is such that there is some k with $(\sigma(k), k) \notin \{(i, \sigma(i))\}$. IE there is an element in the path whose transpose is not in the path. Then the expected value of this path must be 0 since the value at this point is independent and expected to be 0. Therefore we only consider paths whose transpose is themselves and avoid the diagonal (so fixed point free involutions). For each path the expected value of $\prod_i B_{i\sigma(i)}$ is $(-1/2)^n$ since each pair has a 1/2 chance to multiply to 0 and a 1/2 chance to multiply to -1 . The sign of each permutation is -1^n (since there are exactly n swaps). Now we only need to count the number of these such permutations. This is the number of ways to pair up $2n$ numbers. We order the numbers and at each step choose any pair for the lowest unpaired number. The total choices will be

$$(2n-1) \cdot (2n-3) \cdot (2n-5) \cdots 3 \cdot 1 = \frac{(2n)!}{2^n n!}$$

therefore the final answer is:

$$\frac{(2n)!}{2^{2n} n!}$$

13

We might as well consider the entries of the matrix in $\mathbb{F}_2$. Then note that the matrix must have 1s on the diagonal by considering the 1 element subsets. Now we claim the lemma

Claim 2.0.1. A is very odd iff there is a permutation matrix with PAP^{-1} lower triangular.

which very quickly proves the problem (lower triangular matrices with 1s on the diagonal are very odd, and powers preserve this). Now consider what the condition actually means. Make a directed graph whose vertices are the basis vectors with an edge $e_i \rightarrow e_j$ iff $A_{ij} = 1$.

We claim that the lower triangular condition is exactly the statement that the graph has directed cycles of length ≥ 2 (I'll call that a loop). In one direction renumber the indices according to P . Then the lower triangular condition implies that A_{ij} can only be nonzero if $i > j$. This graph can clearly not contain any loops. In the other direction if this graph has no loops choose any ordering of the indices that respects this structure.

Finally, we need to prove that the very odd condition implies that there are no loops. For the sake of contrapositive suppose that there is a minimal nontrivial loop $e_{k_1} \rightarrow e_{k_2} \rightarrow \cdots \rightarrow e_{k_r} \rightarrow e_{k_1}$. Then consider $S = \{k_1, \dots, k_r\}$ and consider the vector $v = e_{k_1} + \cdots + e_{k_r}$. By construction $A_S v = 0$ (since $Ae_{k_i} = e_{k_i} + e_{k_{i+1}}$ on these indices) so the matrix must have zero determinant on these indices.

Eigenvalues and Related Things

14

The answer is yes. Write the system as

$$\frac{d\vec{x}}{dt} = A\vec{x}, \quad A = (a_{ij}), \quad a_{ij} > 0.$$

By Perron–Frobenius, A^T has a real eigenvalue $\lambda > 0$ with an eigenvector

$$v = (v_1, \dots, v_n), \quad v_i > 0, \quad A^T v = \lambda v.$$

Consider the scalar function

$$f(t) := v^T \vec{x}(t) = v_1 x_1(t) + \cdots + v_n x_n(t).$$

Differentiate and use the ODE:

$$f'(t) = v^T \vec{x}'(t) = v^T A \vec{x}(t) = (A^T v)^T \vec{x}(t) = \lambda v^T \vec{x}(t) = \lambda f(t).$$

Hence $f(t) = Ce^{\lambda t}$ for some constant C . Since $x_i(t) \rightarrow 0$ for all i as $t \rightarrow \infty$, we also have $f(t) \rightarrow 0$. But $e^{\lambda t} \rightarrow \infty$, so we must have $C = 0$, and therefore

$$v_1 x_1(t) + \cdots + v_n x_n(t) = 0 \quad \text{for all } t.$$

This is a nontrivial linear relation (since $v \neq 0$), so $x_1, \dots, x_n$ are linearly dependent.

15

This is essentially a problem in representation theory (since the M_i determine a representation of G). It is a corollary of Schur's lemma, but I will not use this machinery for the sake of the context. We will denote:

$$M = \sum_{i=1}^r M_i$$

and note that by the group condition we have

$$M_i M = M = M M_i$$

for each M_i since multiplication by M_i just permutes the indices in the sum. Therefore $M^2 = rM$ or in other words $M' = \frac{1}{r}M$ is a projection matrix. Now M' is a projection with trace 0, a standard fact of linear algebra is that the trace of a projection matrix is the dimension of its image. Now we are done since the image is 0.

16

Since the spectrum is invariant under conjugation we can consider the matrix:

$$\sqrt{B}(A'B)(\sqrt{B})^{-1} = \sqrt{B}A'\sqrt{B}$$

This is a real symmetric matrix, therefore all its eigenvalues are real.

In some sense this is a corollary of Sylvester's law of inertia, where the inertia is defined as the number of positive (resp negative) eigenvalues of A and B . Indeed, a nice generalization is the following (on wikipedia):

Theorem 3.1 (Sylvester's Law of Inertia). If A and B are normal matrices then A is congruent to B (meaning there is some invertible S with $B = S^\dagger A S$) if and only if they have the same number of eigenvalues on each open ray from the origin in the complex plane.

Proof. Note that A defines a conjugate bilinear form:

$$[v_1, v_2]_A = \langle v_1 | A | v_2 \rangle \quad [v_1, v_2]_B = \langle v_1 | B | v_2 \rangle$$

Now consider what happen when we change the basis via a change of basis matrix S . In the new basis we have

$$[v_1, v_2]_A = \langle S v_1 | A | S v_2 \rangle = \langle v_1 | S^\dagger A S | v_2 \rangle$$

so matrices are congruent iff they define the same conjugate bilinear form in different bases. The rest of the proof is left to the reader. $\square$

17

Let $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$ be the characteristic polynomial of A . We take $Af(A)$ and apply Cayley Hamilton and the assumption to yield:

$$A^{n+1} + a_{n-1}A^n + \dots + a_0A = 0 = A^{[n+1]} + a_{n-1}A^{[n]} + \dots + a_0A \quad (1)$$

In particular, each entry of A satisfies something like the characteristic polynomial for A . Now we will induct. Suppose that $A^k = A^{[k]}$ for all $k \leq r$ and $r \geq n+1$. Multiply the LHS of Eq. (1) by A^{r-n} and isolate leading term:

$$A^{r+1} = -a_{n-1}A^r - \dots - a_0A^{r-n+1}$$

In a similar way, we can multiply the RHS of Eq. (1) by $A^{[r-n]}$ "entry wise" to yield:

$$A^{[r+1]} = -a_{n-1}A^{[r]} - \dots - a_0A^{[r-n+1]}$$

By the inductive hypothesis these expressions are equal and $A^{r+1} = A^{[r+1]}$ as desired.

If we initially assume that A and B are invertable we can write $MB^{-1} = A^{-1}M = C$ and:

$$\det(A - MX) = \det(A)^{-1} \det(I - A^{-1}MX) = \frac{\det(I - CX)}{\det(A)}$$

$$\det(B - XM) = \det(B)^{-1} \det(I - XMB^{-1}) = \frac{\det(I - XC)}{\det(B)}$$

Since A and B have the same determinant we need to show that $\det(I - CX) = \det(I - XC)$. This is a corollary of the fact that for any M_1, M_2 the products M_1M_2 and M_2M_1 have the same characteristic polynomial. This can be proven by considering first invertable matrices (here the result follows by conjugating) and then making a continuity argument.

Now we must extend to all matrices. Note that if we replace A and B with $A + \lambda I$ and $B + \lambda I$ they will be invertable for λ outside a certain range. The new A and B still satisfy the properties we used since the condition of having the same characteristic polynomial means that:

$$\det(A + \lambda I) = \det(B + \lambda I)$$

Therefore for λ outside this range write:

$$\det(A + \lambda I - MX) = \det(B + \lambda I - XM)$$

On the other hand both sides are polynomials in λ . Therefore by the identity theorem (complex analysis) they agree for all λ .

Combinatorics

We construct the $n \times n$ adjacency matrix matrix over $\mathbb{F}_2$ as:

$$M_{ij} = \begin{cases} 1, & \text{switches } i \text{ and } j \text{ are connected} \\ 0, & \text{otherwise} \end{cases}$$

And note that is v is a vector with 1s in the location where switches have been flipped, then Mv is the vector with 1s where the light are on. and note that M is a symmetric matrix with 1s on the diagonal. We may also define $w = (1, 1, \dots, 1)$ to be the "all ones vector". We are asked to prove the existence of a $v \in \mathbb{F}_2^n$ with $Mv = w$. Since M is symmetric it suffices to prove that every vector $k \in \ker M$ is orthogonal to w . IE that every vector in the kernel has an even number of 1s. However since M has 1s on the diagonal we may write

$$\begin{aligned} k^T M k &= \sum_{ij} k_i k_j M_{ij} \\ &= \sum_{i \neq j} k_i k_j M_{ij} + \sum_i k_i^2 M_{ii} \\ &= 2 \sum_{i > j} k_i k_j M_{ij} + \sum_i k_i \\ &= \sum_i k_i \end{aligned}$$

where $k = (k_1, \dots, k_n)$. However $k^T M k = k^T (Mk) = 0$ by assumption and we are done.

20

Let $P_{n,k}$ be the number of k -limited permutations of $\{1, 2, \dots, n\}$. Define an $n \times n$ matrix A_n over $\mathbb{F}_2$ where $a_{i,j} = 1$ if $|i - j| \leq k$ and $a_{i,j} = 0$ otherwise. By Leibniz's formula for determinants, $\det(A_n)$ over $\mathbb{F}_2$ evaluates to the parity of $P_{n,k}$.

We want to show $D_n = \det(A_n)$ is periodic with period $2k + 1$. Assume $n \geq 2k + 1$. It will help to draw out the matrices when following this argument. Perform the following operations on A_n over $\mathbb{F}_2$:

- (a) Add column 1 to columns $2, \dots, k + 1$.
- (b) Pass to the minor $A_{1,1}$.

Because we are in $\mathbb{F}_2$, adding row/col 1 cancels out the 1s in the top row (excluding $a_{1,1}$). Expanding the determinant along the first row gives $D_n = \det(M)$, where M is an $(n - 1) \times (n - 1)$ matrix. The top-left $k \times k$ block of M is now strictly zeros.

We can view M as a block matrix:

$$M = \begin{pmatrix} 0 & L & 0 \\ L^T & X & Y \\ 0 & Y^T & A_{n-2k-1} \end{pmatrix}$$

Here, L is a $k \times k$ lower triangular matrix with 1s on the main diagonal, so $\det(L) = 1$. Expanding M along its first k rows isolates the block L . Expanding the complementary minor along its first k columns isolates L^T . Both operations yield a factor of $\det(L) \det(L^T) = 1$.

After evaluating these minors, the remaining submatrix is exactly A_{n-2k-1} . Thus, $D_n = D_{n-2k-1}$, establishing that the sequence is periodic with period $2k + 1$.

Now we need some base cases. For $n \leq k + 1$, A_n is the all-ones matrix J_n . Over $\mathbb{F}_2$, $\det(J_1) = 1$, and $\det(J_n) = 0$ for $n \geq 2$. For $k + 1 < n \leq 2k$, the same matrix operations yield a top-left zero block of size $k \times k$ in an $(n - 1) \times (n - 1)$ matrix. Since $n - 1 < 2k$, this zero block forces the rows to be linearly dependent, so the determinant is 0.

Therefore, the only cases where $D_n \equiv 1 \pmod{2}$ are when $n \equiv 0$ or $1 \pmod{2k + 1}$, completing the proof.

21

Let the vertices be $a_1, \dots, a_n$ and $b_1, \dots, b_n$. Construct the $n \times n$ matrix over $\mathbb{F}_2$ with

$$M_{ij} = \begin{cases} 1, & \text{if } a_i \text{ is connected to } b_j \\ 0, & \text{otherwise} \end{cases}$$

Note the the number of perfect matchings with edge i, j corresponds to looking at the subgraph corresponding to the ij minor of M . Therefore since we are working in $\mathbb{F}_2$ the parity is exactly the determinant of M . We are just asked to find the number of invertible $n \times n$ matrices over $\mathbb{F}_2$. This is

$$\begin{aligned} |GL_n(\mathbb{F}_2)| &= (2^n - 1)(2^n - 2)(2^n - 4) \cdots (2^n - 2^{n-1}) \\ &= \prod_{i=0}^{n-1} (2^n - 2^i) \end{aligned}$$

In general, this is a corollary of the fact the the number of perfect matchings is the permanent of the adjacency matrix, and the permanent agrees with determinant over $\mathbb{F}_2$.

Miscellaneous

22

Consider the linear space of $\mathbb{F}_p[x, y]$ homogenous polynomials with total degree $p - 1$. There is a basis:

$$e_0 = y^{p-1}, e_1 = xy^{p-2}, \dots, e_{p-1} = x^{p-1}$$

Consider the map T with

$$(Tf)(x, y) = f(y, y - x)$$

and consider:

$$Te_i = Tx^i y^{p-i-1} = y^i (y - x)^{p-i-1}$$

The e_j coefficient is

$$\begin{aligned} (-1)^j \binom{p-i+1}{j} &= \frac{(-1)^j}{j!} (p-i-j)(p-i-j+1) \cdots (p-i-1) \\ &= \frac{(-1)^{2j}}{j!} (i+j)(p-i+1) \cdots (p-i-1) \\ &= \binom{i+j}{j} \end{aligned}$$

which proves that T has the same matrix as A . Now we see that

$$(x, y) \mapsto (y, y - x) \mapsto (y - x, -x) \mapsto (-x, -y)$$

however when p is odd this send a homogenous polynomial of degree $p - 1$ to itself as desired. When $p = 2$ this is a special case we can do manually.

23

Note that the determinant of any matrix with row sums equal to 0 is 0, this is because we can add the first $n - 1$ columns to the last column to get a 0 column. Let D_{ij} denotes $\det A[i, j]$. We claim that this implies that $D_{ij} = \pm D_{ik}$. For some matrix A we let $A(x, y) = A + xe_{ij} + ye_{ik}$. Note that $\det A(x, y) = 0$ when $x + y = 0$ by the claim. By expanding minors along the i th row (and noting $\det A = 0$) we see

$$\det A(x, y) = \pm ((-1)^j x D_{ij} + (-1)^k y D_{ik})$$

Therefore plugging in $\det A(x, -x) = 0$ we see that $D_{ij} = (-1)^{j+k} D_{ik}$. Making the symmetrical argument along the columns yields the claim.

Note that the subleading term of the characteristic polynomial of A is explicitly $-\sum_i D_{ii} = -nD$. However this is also the negative of the $n - 1$ symmetric polynomial of eigenvalues. Since the determinant is 0 one eigenvalue, say λ_1 is 0, so this simplifies to

$$\prod_{i=2}^n \lambda_i = nD$$

is the relation.

The type of matrix in the question is related to something called Birkoff polytopes. The solution not really as far as I can tell.

24

Bashing out a bunch of terms reveals the Catalan Numbers.

$$a_n = \frac{1}{n+1} \binom{2n}{n}$$

If we can show that the matrices in the problem statement have determinant 1, then uniqueness follows from expanding out the principle minor in the bottom right corner, and seeing the determinant does nontrivially depend on the next term (with coefficient 1). Define $R(a, b)$ to be the number of paths from $(0, 0)$ with steps $U = (1, 1)$ and $D = (1, -1)$ ending at (a, b) . By "cutting a Catalan path in the middle" we see that

$$C_{a+b} = \sum_{0 \leq k \in \{a, a-2, \dots\}} R(a, k) R(b, k)$$

This is by letting k be the height after a steps, $R(a, k)$ is the ways to get there. The ways to get from (a, k) to $(a+b, 0)$ without going below 0 can be counted by reversing time (starting from the right), therefore it is $R(b, k)$. Therefore define the lower triangular matrices

$$A_{ij} = \begin{cases} R(2i, 2j) & i \geq j \\ 0, & i < j \end{cases} \quad B_{ij} = \begin{cases} R(2i+1, 2j+1) & i \geq j \\ 0, & i < j \end{cases}$$

We see that the first matrix is AA^T and the second is BB^T by our formula for C_{a+b} . However both A and B have 1s on the diagonal. Therefore our desired matrices have determinant 1.

In general, when one has a matrix problem involving combinatorial objects, factoring the matrices somehow is always a good idea. This problem is crazy in that you don't actually need to worry about the formulas for the Catalan or Poll numbers, you only need their properties.

25

If we expand the minors of $V_{n+1}(x)$ along the top row we get two terms (since there are two nonzero things here). The first term is just $xV_n(x)$. The second term we simplify by expanding the minor along the first column (1 term). We get $-V_{n-1}(x)$. Therefore the determinant is:

$$V_{n+1}(x) = xV_n(x) - V_{n-1}(x)$$

26

Recall the recursive formula:

$$V_{n+1}(x) = xV_n(x) - V_{n-1}(x)$$

Since the formula defines $V_n(x)$ exactly it suffices to define

$$f_n(\theta) = \frac{\sin((n+1)\theta)}{\sin(\theta)}$$

and show that

$$f_{n+1}(\theta) = 2 \cos \theta f_n(\theta) - f_{n-1}(\theta)$$

Using $\sin((n+2)\theta) = 2 \cos \theta \sin((n+1)\theta) - \sin(n\theta)$ (product sine formula) and dividing by $\sin \theta$,

$$f_{n+1}(\theta) = 2 \cos \theta f_n(\theta) - f_{n-1}(\theta), \quad f_0(\theta) = 1, \quad f_1(\theta) = 2 \cos \theta.$$

as desired. Since $V_n(x) = \det(xI - A(n))$, the eigenvalues are the roots of V_n . These occur when

$$\sin((n+1)\theta) = 0, \quad \sin \theta \neq 0 \quad \Leftrightarrow \quad \theta = \frac{j\pi}{n+1}, \quad j = 1, \dots, n,$$

so

$$\lambda_j = 2 \cos \left(\frac{j\pi}{n+1} \right), \quad j = 1, \dots, n.$$

27

Consider the circulant matrix in $\mathbb{F}_2$ of the form

$$\begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \end{pmatrix}$$

which has the property that if we plug in v , we get all the indices where the 3 fold sum is odd out. If the matrix is not invertable, the nullity implies that $k = 0$ breaks the condition. Otherwise, there is a bijective mapping, and we are essentially choosing which sums to be 1, so the condition is satisfied. Therefore we are asked to prove for which n is this matrix invertable.

Suppose we want to construct a vector mapping to 0. The condition implies that $v_{n+1} = v_n + v_{n-1}$. We get some cyclic patterns based on the four possible initial choices of v_1, v_2 .

- 000000000...
- 011011011...
- 101101101...
- 110110110...

and the matrix being invertable is exactly that only the first option is possible. However the other 3 options are exactly possible if $n \equiv 0 \pmod{3}$ since they all cycle with period 3. Therefore the problem condition is satisfied iff $n \not\equiv 0 \pmod{3}$.

28

This problem seems to be very connected to Clifford algebras. Sorry, I'll come back to this when I've read more about them (hopefully soon!) and can give a proof that I'm proud of. It is a bit sad that there is one problem I haven't written anything on (ruining my completion). However reading more math is always good.

29

(a) Let M be the matrix in the problem. For any non-zero vector $v \in \mathbb{R}^n$, we consider the quadratic form:

$$v^T M v = \sum_{i=1}^n \sum_{j=1}^n x_i x_j \frac{1}{(i+j)^2}$$

Using the integral identity $\int_0^\infty t e^{-(i+j)t} dt = \frac{1}{(i+j)^2}$, we substitute and rearrange:

$$\begin{aligned} v^T M v &= \sum_{i,j} x_i x_j \int_0^\infty t e^{-it} e^{-jt} dt \\ &= \int_0^\infty t \left(\sum_{i=1}^n x_i e^{-it} \right) \left(\sum_{j=1}^n x_j e^{-jt} \right) dt \\ &= \int_0^\infty t \left(\sum_{k=1}^n x_k e^{-kt} \right)^2 dt \end{aligned}$$

Since $t > 0$ and the squared term is non-negative, the integral is non-negative. For the integral to be zero, the function $f(t) = \sum_{k=1}^n x_k (e^{-t})^k$ must be zero for all $t \in (0, \infty)$. Let $u = e^{-t}$; then $P(u) = \sum_{k=1}^n x_k u^k$ is a polynomial that must vanish on the interval $(0, 1)$. By the fundamental theorem of algebra, this implies all coefficients $x_k = 0$.

Thus, for $v \neq 0$, we have $v^T M v > 0$, proving M is positive definite.

(b) An approximation of the integral via discrete sums is given by an $n \times n$ matrix equation:

$$v^T A v \quad v_i = f(a + (b-a)i/n) \quad A_{ij} = \frac{1}{n^2} \frac{1}{(2a + \frac{b-a}{n}(i+j))^2}$$

Note that A can be shown to be positive definite by the same argument as (a). As $n \rightarrow \infty$ this approaches the integral, however by since A is positive definite it is positive.

30

We claim by induction that the eigenvalues of A_n form an arithmetic sequence $r_0, \dots, r_n$ with common ration 2 centered around 0 (so that $r_0 = -r_n$). Moreover the multiplicity of r_i is $\binom{n}{i}$ with that many distinct r_i -eigenvectors v_{ik} . The base case is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ having eigenvectors $1, -1$ with eigenvectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Proof. If we can construct that many distinct eigenvectors for each eigenvalue we are done by dimension counting. We do this by induction. Note that each eigenvector v_{ik} gives new eigenvectors.

$$\begin{bmatrix} v_{ik} \\ v_{ik} \end{bmatrix} \quad \begin{bmatrix} v_{ik} \\ -v_{ik} \end{bmatrix}$$

with eigenvalues $r_i + 1$ and $r_i - 1$ respectively. Moreover all the eigenvectors constructed this way are linearly independent by construction. This gives exactly the recurrence relation for the binomial coefficients and we are done. $\square$

31

We might as well replace f with $zf - z^n + 1$ (roots shared are still the same). We will denote c_i as the i coefficient to this new polynomial so $f(z) = \sum_{j=0}^{n-1} c_j z^j$ (constant term $c_0 = 1$). The roots of $z^n - 1$ are exactly ω^r where ω is an n th root of unity. This is essentially a fourier analysis problem. In the contrapositive of the problem statement, suppose that f is supported on less than $\sqrt{n}$ of the n th roots of unity. Set

$$b_k := f(\omega^k) \quad (k \in \mathbb{Z}/n\mathbb{Z}), \quad S := \{k : b_k \neq 0\}.$$

It is general there is the very illuminating equivalence

$$\begin{aligned} \text{Multiplication/Convolution in Time} &\iff \text{Pointwise Multiplication in Frequency} \\ \text{Multiplication/Convolution in Frequency} &\iff \text{Pointwise Multiplication in Time} \end{aligned}$$

and vice versa. This is the Discrete Fourier situation. Let $g(x) = \sum_{j=0}^{n-1} c_{n-j} z^j$ be the time reversed f with fourier coefficients $d_k = c_{n-k}$. Now consider convolving f and g in the Frequency domain

$$(B * D)_s = \sum_k b_k d_{s-k} = \sum_k b_k b_{k-s}$$

but by the Fourier convolution theorem this is

$$(B * D)_s = n \sum_{k=0}^n c_n c_{n-j} \omega^{sj}$$

By the hypothesis $c_j c_{n-j} = 0$ for $1 \leq j \leq n-1$, so only the 0 term contributes, it contributes $nc_0^2 = n$.

Since $(B * D)_s \neq 0$, for each s there exists some k with $b_k b_{k-s} \neq 0$, hence $k \in S$ and $k-s \in S$. Therefore $s = k - (k-s) \in S - S$. So $S - S = \mathbb{Z}/n\mathbb{Z}$ and

$$n = |S - S| \leq |S|^2.$$

Hence $|S| \geq \sqrt{n}$, i.e. at least $\sqrt{n}$ of the values $f(\omega^k)$ are nonzero.

32

Note that P is reflection about the orthogonal plane to u . Therefore it is a unitary matrix and PQ is unitary. Since $-Q$ is unitary, it's diagonalizable with roots conjugate pairs. Since each conjugate pair multiplies to 1, the determinant of $-Q$ is $(-1)^{\text{mult of eigenvalue } -1}$. If it has no -1 eigenvalues, the determinant is 1, but then the determinant of $(-Q)P$ is -1 and it must have at least one -1 eigenvalue. Therefore if Q has no 1 eigenvalues then QP has at least one.

This is an old Putnam problem I did maybe two years ago, and I remember there was an official solution using the Cayley transformation. I might work that out here later if I become inclined.